$\Sigma = \{a . b . c\}$ $\{0.1\}$ $L \leq \Sigma^*$

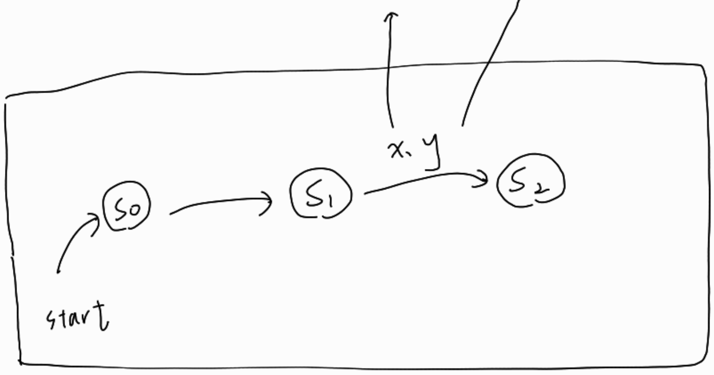
Σ^2 $L = \{0.1,\ 00.\ 01\}$

regular expression

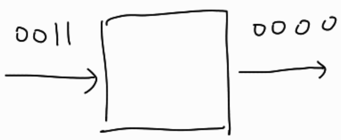
ex: $0 \cdot 1\ \{0.1\}^{*}_{}{}^{+}$

F.S.M

$M = (S.\ I.\ O.\ V.\ W)$



M: string recognizer



001110
000010

00111
00001

yes , No

$\{0.1\}^* 111$ RE FSM

$(\ ) \in L$

$(x) \in L$ if $x \in L$

$xy \in L$ if $x.y \in L$

We cannot construct a F.S.M recognize $A = \{0.1 \mid i \in Z^+\}$

$|S| = n$

$W(S_0, 01) = 01$

$W(S_0, 0011) = 0011$ $W(0^{n+1} 1^{n+1}) = 0 \cdots\cdots 1$

$M = (S.\ I.\ O.\ V.\ W)$

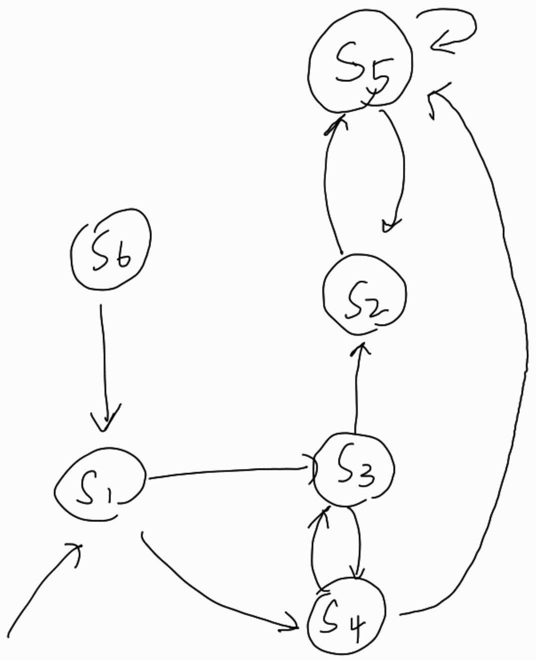
state	S_0	S_1	S_2					
I	0	0	0	0	0	0	1	1
O	0					0		1

n+1 n+1

F. S. M. minimization

$S_1 \, E_{k+1} \, S_2$ if $v(S_1, X) \; E_k \; v(S_2, X)$ for $X \in I$

Ex:



	v: 0	v: 1	w: 0	w: 1
S_1	S_4	S_3	0	1
S_2	S_5	S_2	1	0
S_3	S_2	S_4	0	0
S_4	S_5	S_3	0	0
S_5	S_2	S_5	1	0
S_6	S_1	S_6	1	0

$P_0 \; \{S_1, S_2 \; \ldots \; S_6\}$

$P_1 \; \{S_1\} \; \{S_3, S_4\} \; \{S_2, S_5, S_6\}$

$P_2 \; \{S_1\} \; \{S_3, S_4\}$ $\quad P_3 \; \{S_1\} \; \{S_3 \; S_4\}$

$v(S_2, 0) = S_5$

$v(S_6, 0) = S_1$

$S_3 E_1 S_4 \quad S_3 E_2 S_4$

$v(S_3, 0) = S_2$

$v(S_4, 0) = S_5$ $\quad S_2 E_1 S_5$

$v(S_3, 1) = S_4$

$v(S_4, 1) = S_3$ $\quad S_4 E_1 S_3$

$\therefore \; S_3 E_2 S_4$